

Automatic radiology report generation: a systematic review of emerging AI architectures and multimodal technologies

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Automatic radiology report generation: A Systematic Review of Emerging AI Architectures and Multimodal Technologies

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Abstract

Generating radiology reports from medical images is a labour-intensive task, exacerbated by the global shortage of radiologists and the complexity of image interpretation. Automatic radiology report generation (ARRG) has emerged to address this challenge, with AI-driven methods showing promise in accelerating diagnosis and improving workflow efficiency. While previous reviews primarily focus on conventional AI and deep learning models and often adopt non-systematic methodologies, this work provides a systematic PRISMA-based review of 51 studies published between 2015 and 2025, offering a rigorous synthesis of model architectures, learning paradigms, data strategies, and evaluation metrics. We categorize approaches from classical CNN-RNN frameworks to emerging paradigms including knowledge graphs, large language models (LLMs), retrieval-augmented generation (RAG), AI agents, and multimodal vision-language models. Our analysis highlights key challenges in clinical translation, including model explainability, factual accuracy, generalization across modalities, and deployment feasibility. We also critically evaluate benchmark datasets and performance metrics, identifying inconsistencies that hinder fair comparison and clinical adoption. Finally, we propose actionable research directions to advance clinically reliable, explainable, and generalizable ARRG systems. By linking technical developments to real-world clinical requirements, this review serves as a forward-looking reference for researchers and practitioners aiming to bridge the gap from AI prototypes to trusted clinical tools.

Keywords

Automatic Radiology Report Generation (ARRG), Deep Learning Algorithms, Neural Networks, Graph Neural Networks (GNN), Large Language Models (LLMs).

Introduction

The accurate handling of medical data is a fundamental requirement in healthcare, where diagnostic decisions directly affect patient outcomes. This need for precision, combined with the increasing complexity of clinical data, has encouraged the adoption of computational technologies, particularly machine learning (ML) and deep learning (DL), to support clinical decision-making. In medical imaging, these techniques have demonstrated considerable potential to enhance diagnostic accuracy, improve workflow efficiency, and reduce clinician workload, positioning radiology as a key domain for artificial intelligence (AI) driven innovation. Radiological imaging modalities such as, X-ray, computed tomography (CT), and magnetic resonance imaging (MRI) provide essential, non-invasive insights into anatomical structures and pathological conditions. The interpretation of these examinations culminates in radiology reports, which serve as a primary communication medium between radiologists and other healthcare professionals. These reports are expected to be clear, accurate, concise, complete, and clinically consistent. However, despite their critical role, radiology reports lack a universally standardized structure, with reporting practices varying across institutions and countries. Traditional, manually generated reports therefore remain susceptible to variability in style, content, and interpretation, as well as to human error. Systemic pressures within modern healthcare systems further amplify the limitations of conventional reporting practices. The growing demand for imaging examinations continues to outpace the expansion of the radiology workforce. According to the World Health Organization (2024), the global shortage of radiologists is estimated to range between 30,000 and 50,000 professionals, with particularly severe consequences in underserved and rural regions. Imaging volumes increase annually by approximately 8-12%, while workforce growth remains limited to 2-3% (European Society of Radiology, 2025)¹. As a result, radiologists are required to interpret between 50 and 100 cases per day, dedicating nearly 20–30% of their time to report writing (American College of Radiology, 2025)². Under these conditions, diagnostic error rates persist at 3-5%, and report turnaround times can extend to several days, delaying patient care. These challenges have motivated sustained research efforts toward automating or assisting the radiology reporting process. Beyond addressing workforce shortages, automated solutions aim to reduce inter- and intra-observer variability, improve report consistency, and enable more timely clinical decision-making. Early attempts to support radiologists relied on classical machine learning approaches and rule-based systems, which provided limited scalability and adaptability. The evolution of data availability and computational power subsequently enabled the development of more advanced AI-driven solutions. The emergence of Computer-Aided Diagnosis (CAD) systems marked an important milestone in this evolution. Beginning in the 1980s with lesion detection in chest radiographs and mammography, CAD systems gradually became integrated into clinical workflows, particularly in mammography during the early 2000s, where they improved radiologist sensitivity by 5-10% and reduced false-negative rates. While early CAD systems focused primarily on detection and classification tasks, recent advances have extended their scope toward higher-level clinical reasoning and automated report generation.

Automatic Radiology Report Generation (ARRG) represents a natural progression of this paradigm. (Shin et al. 2015), who employed Convolutional Neural Networks (CNNs) to generate keywords from CT and MRI images, presented one of the earliest contributions in this area. This work was extended by (Shin et al. 2016), who proposed a framework capable of detecting diseases and describing their clinical context using chest X-ray images and corresponding reports. Since 2018, research activity in ARRG has increased substantially (Xue et al. 2018; Li et al. 2018; 2019), with studies exploring a wide range of deep learning architectures and multimodal learning strategies. Modern ARRG systems increasingly rely on sophisticated neural architectures, including CNNs for visual feature extraction, Recurrent Neural Networks (RNNs) for sequential modelling, and Transformer-based models for capturing long-range dependencies across multimodal inputs. More recently, Large Language Models (LLMs), such as GPT-4, have been investigated for their ability to generate fluent and context-aware radiology reports. To address known limitations such as hallucinations and factual inconsistency, Retrieval-Augmented Generation (RAG) techniques have been introduced, while Graph Neural Networks (GNNs) have been explored

¹ <https://www.myesr.org/>

² <https://www.acr.org/>

for modelling complex relational structures within medical data. These components collectively form the technological foundation of contemporary ARRГ systems, aiming to balance automation efficiency with clinical reliability and explainability. Several comprehensive review studies have examined the rapid development of ARRГ methodologies, summarizing prevailing architectures, datasets, evaluation strategies, and open challenges (Allaouzi et al. 2018; Monshi et al. 2020; Liao et al. 2023; C. Pereira et al. 2024; Liu et al. 2025). Other surveys focus on specific imaging modalities, most notably chest radiography, where ARRГ systems are particularly mature (Kaur et al. 2022). Despite promising results, recent systems such as (Arasteh et al. 2025) indicate that while diagnostic accuracy can be improved by up to 54%, challenges related to false positives, explainability, and clinical trust remain unresolved. A comparative overview of traditional and AI-based radiology report generation approaches is presented in Table 1, highlighting differences in workflow, efficiency, data requirements, regulatory considerations, and ethical implications.

Aspect	Traditional-Based Methods	AI-Based Methods
Process	Manual interpretation and report writing by radiologists	Automated or semi-automated report generation using AI algorithms
Characteristics	Time-consuming; relies heavily on radiologist's expertise; prone to variability	Rapid processing; utilizes machine learning and deep learning; consistent output
Advantages	High interpretability; personalized insights; established clinical trust	Increased efficiency; ability to handle large datasets; potential for improved diagnostic accuracy
Limitations	Subject to human error; variability in reports; delays in report turnaround	Requires large, annotated datasets; potential for algorithmic bias; challenges in explainability and transparency
Data Requirements	Minimal digital infrastructure; relies on radiologist's knowledge	Necessitates extensive digital infrastructure; dependent on quality and diversity of training data
Regulatory Aspects	Well-established guidelines and standards	Emerging regulatory frameworks; concerns over validation and approval processes
Ethical Considerations	Maintains patient-doctor relationship; direct accountability	Raises questions about data privacy, consent, and the extent of AI involvement in clinical decision-making

Table 1: Traditional-Based vs. AI-Based Radiology Report Generation Characteristics

While AI-based systems demonstrate clear advantages in scalability and consistency, their clinical adoption remains limited. Recent surveys indicate that although approximately 30% of radiology departments in developed countries employ at least one AI-based tool, fewer than 5% have adopted ARRГ systems (RSNA AI Survey, 2025). This underutilization persists despite the rapid growth of the medical AI market, which is projected to expand from \$15.1 billion to \$187.9 billion by 2030 (Grand View Research, 2025). Handling medical data with precision is crucial due to the rigorous and complex nature of the field. This has driven the adoption of advanced technologies, particularly machine learning (ML) algorithms, in aiming to develop ML tools especially deep learning (DL) models, which shows great promise in improving diagnostic accuracy and treatment outcomes.

Research Gap and Motivation

Despite rapid advances in Automatic Radiology Report Generation (ARRГ) and the growing clinical pressure caused by radiologist shortages, existing ARRГ research remains largely misaligned with real-world clinical requirements. As outlined in the preceding section, most current systems demonstrate promising technical performance but fall short of providing clinically reliable, explainable, and deployable solutions. A primary gap lies in evaluation practices. The majority of ARRГ studies and surveys rely on linguistic similarity metrics such as BLEU, ROUGE, and CIDEr, which inadequately reflect diagnostic correctness and clinical utility. As demonstrated by (Yu et al. 2023), these metrics show only moderate correlation ($R^2 = 0.35-0.48$) with clinically grounded measures such as RadGraph F1, indicating that fluent report generation does not necessarily translate into accurate or actionable clinical information. This misalignment has led to optimization toward benchmark scores rather than patient-centered outcomes. In parallel, critical aspects required for clinical translation remain underrepresented in existing surveys. Regulatory readiness, including compliance with FDA 510(k) and EU MDR pathways, risk management standards such as ISO 14971, and algorithmic bias assessment, is rarely discussed. Similarly, explainability is often limited to attention mechanisms or visualization techniques that provide localization without causal reasoning, despite evidence that robust explainability can significantly improve radiologist trust (Borys et al. 2023) and remains largely experimental in advanced forms (Alam et al. 2025).

Paper	Type of Review	ARRG Papers Reviewed	Scope & Focus	Contribution	Strengths	Weaknesses	Datasets & Benchmarks	Evaluation Metrics	Clinical Relevance discussed
(Jing et al. 2018a)	Narrative Review	~15	CNN-RNN architecture for report generation	Pioneering deep learning-based framework for report generation	Defined benchmarks, evaluation baselines	Early models only; outdated	IU-XRay; Partial benchmarks	BLEU, METEOR, CIDEr	Partial
(Monshi et al. 2020)	Taxonomy Paper	~105	Categorization of CNN-RNN-based systems	First comprehensive review on DL models for ARRG; categorized CNN-RNN pipelines and evaluation metrics	Solid foundation on CNN-RNN methods, datasets, and metrics	Lacks coverage of transformer-based and multimodal models	IU-XRay, MIMIC-CXR; Yes	BLEU, ROUGE, METEOR	No
(Messina et al. 2020)	Narrative Review	~167	Explainability and clinical NLP in radiology	Introduced explainability and interpretability concerns; emphasized clinical relevance	Covers both algorithmic and interpretability aspects	Overlaps with general explainable AI; less focus on ARRG-specific design	MIMIC-CXR; No benchmarks	None	Yes
(Pavlopoulos et al. 2022)	Systematic Review	~65	Text-only NLP tools in radiology (non-vision)	Positioned diagnostic captioning as a subfield of ARRG; introduced linguistic and	Detailed analysis of captioning errors and challenges	Focused more on image captioning than full report generation	OpenI, RadLex; No benchmarks	NLP F1, precision/recall	Yes

				clinical error taxonomies					
(C. Liu et al. 2024)	Taxonomy Paper	41	DL model evolution (RNNs to transformers)	Systematic PRISMA-based review; synthesized performance trends; analyzed design patterns	Methodical literature selection and synthesis	Misses 2022+ developments (transformer, LLM)	IU-XRay, MIMIC-CXR; Benchmarks	BLEU, CIDEr, ROUGE	No
(Sloan et al. 2024)	Systematic Review (PRISMA)	~84	Comprehensive DL report generation pipeline	Surveyed cutting-edge ARRG models (transformers, co-attention, LLMs); outlined open research challenges	Up-to-date, discusses multi-section generation	Brief treatment of evaluation and clinical deployment	IU-XRay, MIMIC-CXR, ChestX-ray14; Full	BLEU, CIDEr, CheXbert, ROUGE, BERTScore	Yes
(X. Wang et al. 2024)	Narrative Review	Not specified	LLMs and transformers in ARRG	Focused on fusion of image + text + structured data; suggested benchmark framework	Emphasizes real-world multimodal workflows	Lacks focus on generation quality/error analysis	MIMIC-CXR, MS-CXR; Benchmarks	BLEU, RadGraph F1, BERTScore	Partial
(Yi et al. 2025)	Narrative Review	Not specified	Multimodal (text, image, metadata) integration	First survey focusing on LLMs (e.g., GPT-4V, Med-PaLM) for ARRG + VQA; explores model architectures	Pioneering discussion of LLM integration in radiology	Preliminary - limited empirical benchmarking	MIMIC-CXR, OpenI; Minimal	Sparse metric use	Yes

(Liu et al. 2025)	Taxonomy + Narrative	~40	Multimodal LLMs for reporting + VQA	Unified view of multimodal and LLM-based ARRГ	Combines recent advances into unified survey	Limited real-world validation	MIMIC-CXR, MS-CXR, PMC-VQA; Benchmarks	BLEU, BERTScore, RadGraph, VQA acc.	Yes
Our Work	Systematic Review	51	Comprehensive automatic radiology report generation pipeline covering traditional, deep learning, multimodal, and LLM-based methods	Unified view of ARRГ methods; critical analysis of clinical translation, explainability, deployment, and regulatory readiness; identifies gaps and future directions	Integrates recent advances (CNNs, RNNs, Transformers, LLMs, RAG, GNNs); explicit clinical translation lens; compares datasets and evaluation metrics; highlights practical deployment challenges	Covers a large scope, which may make some sections dense	IU X-ray, ChestX-ray, CheXpert, PadChest, VinDr-CXR, MIMIC-IV	BLEU, ROUGE, CIDEr, RadGraph, BERTScore; analysis of alignment with clinical correctness and radiologist trust	Yes – explicitly evaluated along diagnostic accuracy, explainability, scalability, and regulatory considerations

Table 2: Comparative Analysis of Existing Review Papers on ARRГ

Furthermore, most ARRГ systems are evaluated on single-institution datasets, leading to substantial performance degradation (15–30%) when deployed across heterogeneous clinical settings (Table 2). Practical deployment challenges such as PACS/RIS³ integration, privacy compliance, real-time performance, and workflow adaptation are rarely examined. Although recent studies have begun to address radiologist preferences (Xiao et al. 2024), multimodal semantic modelling (Cheddi et al. 2024), and clinically informed evaluation metrics (Luping 2024), these efforts remain fragmented and lack a unified, clinically oriented synthesis. As summarized in Table 2, existing ARRГ reviews either emphasize early template-based systems or focus narrowly on deep learning approaches for chest X-rays. Few offer an integrative, cross-modality perspective or critically analyse evaluation metrics in relation to clinical relevance. To address these shortcomings, our review first synthesizes the main research directions established by early and contemporary work in the field, and then focuses on the technologies that have demonstrated practical value for automatic radiological report generation. Taken together, these limitations reveal a clear gap between algorithmic innovation and clinical applicability. This gap motivates the need for a comprehensive ARRГ survey that systematically integrates technical advances with evaluation rigor, explainability, regulatory considerations, and deployment feasibility, thereby supporting the translation of ARRГ systems from experimental settings into routine clinical practice.

Paper Contributions

Motivated by these gaps, this paper aims to provide a comprehensive and clinically oriented survey of Automatic Radiology Report Generation systems. Unlike existing reviews that focus predominantly on model architectures and benchmark results, this work explicitly examines the readiness of ARRГ systems for clinical translation, emphasizing validation, explainability, regulatory considerations, and deployment feasibility.

The main contributions of this paper are as follows:

- ✓ Provide a comprehensive overview of the state of the art in ARRГ systems, covering classical and modern AI-based approaches.
- ✓ Analyse and compare existing frameworks, architectures, and enhancement strategies with explicit attention to clinical accuracy, explainability, scalability, and robustness.
- ✓ Systematically compare this survey with previous ARRГ review papers through the lens of clinical translation readiness.
- ✓ Review the most relevant datasets and evaluation metrics used in current ARRГ research and critically assess their clinical relevance.
- ✓ Examine recent innovations, including Large Language Models (LLMs), retrieval-augmented generation (RAG), AI agents, and multimodal deep learning, with a focus on deployment feasibility.
- ✓ Identify key barriers to clinical adoption and propose actionable pathways for regulatory approval and real-world implementation.
- ✓ Evaluate explainability mechanisms in relation to radiologist trust and clinical decision support requirements.
- ✓ Assess multi-institutional generalization challenges and discuss standardization strategies for broader clinical deployment.

Methodology

This study adopts a hybrid review methodology that integrates a PRISMA-compliant systematic review with a theory-driven narrative synthesis, enabling both methodological rigor and in-depth analytical insight. This design is increasingly adopted in high-impact medical imaging reviews to move beyond descriptive comparisons and support structured interpretation of heterogeneous AI methodologies. The systematic component ensures transparency, reproducibility, and comprehensive coverage of the literature, while the narrative synthesis facilitates critical analysis of methodological trends, technological evolution, and clinical relevance in Automatic

³ Picture Archiving and Communication System/ Radiology Information System

Radiology Report Generation (ARRG). The overall study selection process is illustrated in Figure 1, and eligibility criteria are summarized in Table 3.

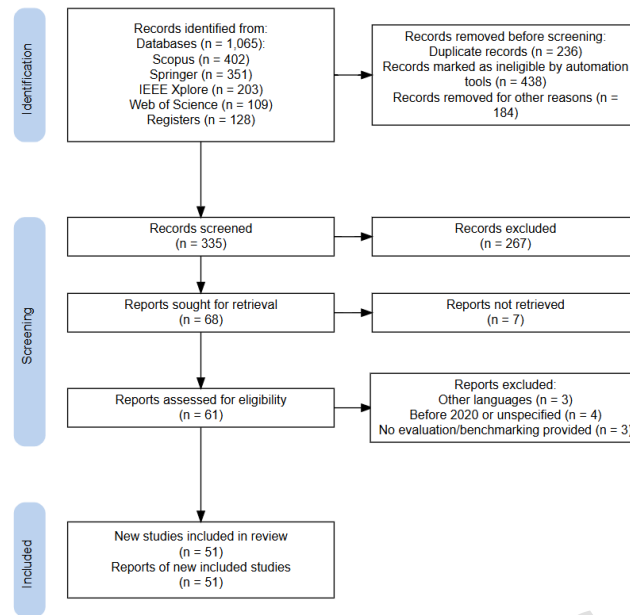


Figure 1: PRISMA schema for papers selection

Criteria	Inclusion	Exclusion
Domain	Automatic radiology report generation	Non-radiology or manual systems
Language	English	Other languages
Publication Date	2010–2025	unspecified
Technologies	Machine Learning, Deep learning techniques	Classics algorithms
Evaluation	Includes performance evaluation	No evaluation/benchmarking provided

Table 3: Eligibility Criteria

Eligibility Criteria

Studies were selected based on predefined inclusion and exclusion criteria designed to ensure relevance, methodological quality, and clinical significance (Table 3). Eligible publications were required to focus explicitly on automatic radiology report generation and to employ machine learning or deep learning-based techniques. To ensure methodological comparability, only studies reporting quantitative performance evaluation were included. Non-radiological applications, manual reporting systems, classical rule-based or non-learning algorithms, and studies lacking experimental validation were excluded. Only English-language publications released between 2010 and 2025 were considered, reflecting the emergence and maturation of data-driven ARRG systems during this period.

Data Sources and Search Strategy

A comprehensive literature search was conducted across major medical and technical databases to ensure broad and balanced coverage. The following repositories were systematically queried: ScienceDirect, IEEE Xplore, SpringerLink, Scopus, Web of Science, PubMed, and arXiv. This selection captures peer-reviewed journal articles,

high-impact conference proceedings, and influential preprints at the intersection of medical imaging and artificial intelligence. Search queries were constructed using combinations of controlled vocabulary and free-text terms related to radiology report generation, medical image captioning, vision-language models, deep learning, and generative AI. Particular emphasis was placed on recent advances in deep learning and multimodal architectures, reflecting their dominant role in contemporary ARRГ research.

Study Selection Process

The study selection followed a multi-stage screening process in accordance with PRISMA guidelines. After removing duplicates, titles and abstracts were screened for relevance, followed by full-text assessment against the eligibility criteria. Discrepancies during screening were resolved through iterative review and consensus. The final set of included studies forms the basis for qualitative synthesis and comparative analysis.

Analysis Framework: A Clinical Translation Lens

To distinguish this review from prior ARRГ surveys, we adopt a clinical translation oriented analytical framework that evaluates ARRГ systems beyond benchmark accuracy. Rather than treating report generation as a purely algorithmic task, each approach is assessed in terms of its readiness for real-world clinical deployment.

Specifically, the reviewed studies are analysed across four clinically relevant dimensions:

1. Diagnostic accuracy and patient safety, focusing on the clinical validity of generated reports and the potential consequences of diagnostic errors.
2. Explainability and interpretability, examining whether models provide transparent, clinically meaningful rationales that support radiologist trust and accountability.
3. Scalability and deployment feasibility, including computational efficiency, integration with existing clinical systems (e.g., PACS/RIS), and operational constraints.
4. Validation and regulatory considerations, addressing external validation practices, risk mitigation strategies, and alignment with regulatory pathways such as FDA 510(k) clearance and EU MDR compliance.

This framework enables a structured comparison of ARRГ methodologies from both technical and clinical perspectives, revealing gaps between research performance and clinical applicability. By embedding translational criteria directly into the methodological analysis, this review provides a principled foundation for evaluating not only how ARRГ systems perform, but also whether they are suitable for routine clinical use.

ARRГ System Architecture: A Pipeline-Based Taxonomy & Clinical Analysis

Overview of the ARRГ Pipeline

Automatic Radiology Report Generation (ARRГ) systems are commonly organized as a three-stage pipeline (Figure 2), consisting of feature extraction, clinical reasoning and report drafting, and report refinement and explainability. Architectural decisions at each stage play a critical role in determining clinical accuracy, interpretability, and scalability, and therefore directly affect real-world clinical utility (Hou et al. 2023; Xue et al. 2022; M. Li, Liu, et al. 2023). As summarized in Table 4, the progression of ARRГ architectures reflects growing requirements for diagnostic precision, linguistic coherence, and clinical relevance. Early systems were predominantly based on retrieval strategies or simple CNN-RNN and sequence-to-sequence frameworks, where convolutional neural networks (CNNs) extracted visual features and recurrent neural networks (RNNs) generated textual reports. While effective for structured and template-driven reporting, these approaches exhibited limited semantic flexibility and struggled to capture complex clinical context.

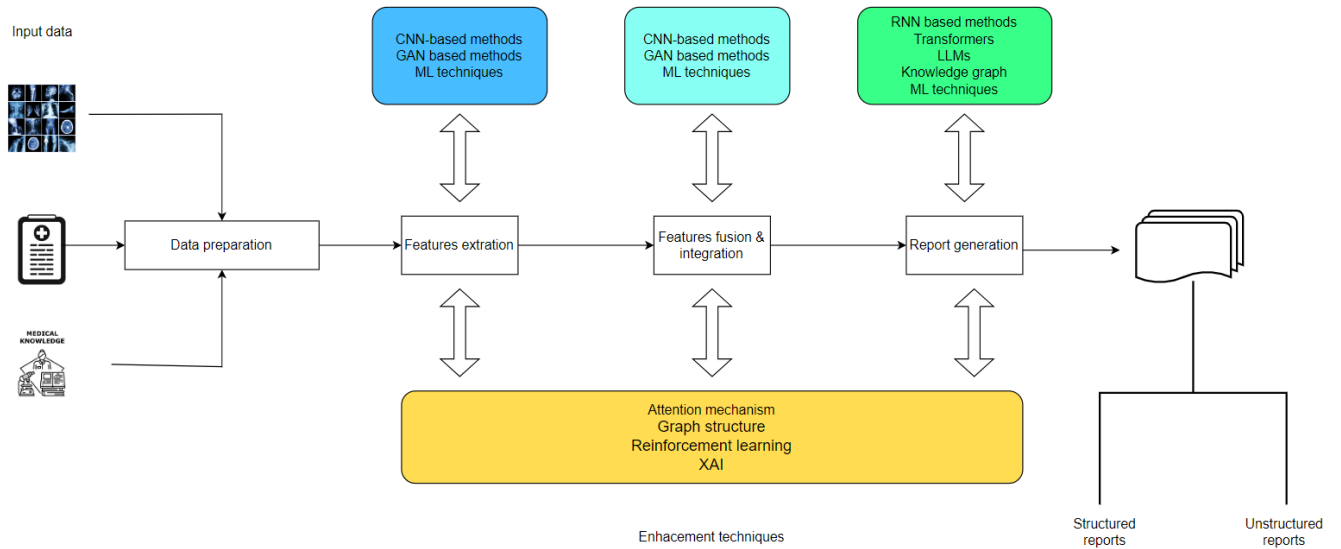


Figure 2: ARR system workflow

More recent ARR systems have moved toward deeper and more expressive paradigms, including multimodal deep learning, attention-based mechanisms, knowledge-enhanced models, and large language model-driven architectures. These advances aim to improve lesion sensitivity, contextual reasoning, and report coherence while supporting richer clinical semantics. Deep learning methods particularly CNNs, RNNs, generative adversarial networks (GANs), and retrieval-augmented generation (RAG) frameworks now constitute the dominant methodological foundation of ARR research.

Architecture	Clinical Accuracy Driver	Explainability Potential	Scalability & Deployment Consideration	Typical Use in ARR
CNN-RNN (+Attention)	Attention mechanisms align salient image regions with descriptive text, improving finding description.	Visual attention maps provide intuitive, region-level justification for generated text.	Simple architectures are computationally lightweight but sequential generation limits inference speed.	Foundational framework for early deep learning-based ARR (Jing et al. 2018b).
Transformer-based	Self-attention captures long-range dependencies within the image and report, improving coherence for multi-sentence narratives.	Self-attention weights are complex and high-dimensional, posing challenges for human interpretation.	Parallelizable training is efficient, but large parameter counts demand significant resources.	Becoming the standard encoder-decoder framework for state-of-the-art models (Chen et al. 2020).
LLM-based (Fine-tuned)	Leverages vast pre-trained biomedical knowledge for superior semantic understanding and terminology use.	Can be guided via prompt engineering (e.g., chain-of-thought) to output intermediate reasoning steps.	High resource demands for training and inference; cost and latency are key deployment hurdles.	Used as powerful decoders in multimodal systems (e.g., LLaVA-Med, Med-Gemini).
Retrieval-Augmented (RAG)	Grounds generation in external, verifiable knowledge bases (e.g., RadLex, PubMed), drastically improving	Retrieved references offer explicit, citable evidence for generated content, enhancing trust.	Introduces latency from the retrieval step; requires maintenance of curated, up-to-date knowledge bases.	Used to enhance factuality in systems like RadioRAG (Arasteh et al. 2025) and MMed-RAG (Xia et al. 2025).

	factuality and reducing hallucination.			
Graph-enhanced (KG/GNN)	Explicitly models relationships between anatomical structures and findings, mimicking clinical reasoning for more logical report flow.	The graph structure itself is an interpretable representation of the model's "understanding" of medical concepts.	Construction of high-quality knowledge graphs is resource-intensive; GNNs add computational overhead.	Integrates prior knowledge to improve report semantics (Y. Zhang et al. 2020; Y. Liu et al. 2024).

Table 4: Core ARRГ Architectures and Their Clinical Implications

In this section, ARRГ architectures are analysed from an ARRГ-specific clinical perspective, rather than through generic machine learning characteristics. We examine (Huang et al. 2017; He et al. 2015; M. Li, Lin, et al. 2023; Li et al. 2024) how feature extraction strategies influence diagnostic sensitivity, how reasoning and drafting mechanisms affect report consistency and terminology alignment, and how refinement and explainability components contribute to transparency and clinician trust. The clinical implications of these architectural choices are consolidated in Table 4 to facilitate systematic comparison across ARRГ paradigms.

Feature Extraction, fusion, and Representation for Radiology Report Generation

Extracting clinically meaningful visual features is the foundation of Automated Radiology Report Generation (ARRГ). Unlike natural image captioning, radiology requires sensitivity to subtle, multi-scale abnormalities such as ground-glass opacities, small nodules, or fractures where missed findings can directly affect diagnosis and patient outcomes. Consequently, feature extraction in ARRГ is not a generic vision task but a clinically constrained design problem.

From Generic CNNs to Radiology-Optimized Encoders

Convolutional neural networks remain the dominant visual encoders in ARRГ; however, architectures pre-trained on natural images exhibit limitations when transferred directly to medical imaging. Radiology-specific encoders are required to preserve fine-grained pathological cues while maintaining robustness across imaging protocols and patient populations. DenseNet architectures (DenseNet-121/169) are the most widely adopted in ARRГ systems (Rajpurkar et al. 2017; Xue et al. 2018; Y. Liu et al. 2024). Their dense connectivity facilitates feature reuse and improves sensitivity to small or low-contrast abnormalities. In contrast, ResNet-based encoders (ResNet-50/101), while computationally efficient, may underrepresent subtle findings unless augmented with attention mechanisms (Yi et al. 2024; Hou et al. 2023). Comparative studies report superior clinical label performance for DenseNet, including higher AUC for pleural effusion detection (0.87 vs. 0.83). More recently, Vision Transformers (ViT) have been explored for ARRГ (Dosovitskiy et al. 2020; Zhang and Jiang 2024). By modeling global spatial dependencies, ViTs are well suited for detecting diffuse or distributed pathologies. However, their reliance on large training datasets and increased computational cost raises scalability concerns for routine clinical deployment.

Clinical Accuracy Implications of Encoder Choice

Encoder selection has direct consequences for diagnostic reliability (Liao et al. 2023). Architectures that preserve high-resolution and multi-scale features demonstrate improved sensitivity to clinically relevant findings, while overly compressed representations increase the risk of false negatives. These trade-offs underscore the importance of aligning encoder design with clinical priorities rather than solely computational efficiency.

Explainability and Clinical Trust in Feature Extraction

Explainability at the feature extraction stage is a key determinant of clinical trust in Automatic Radiology Report Generation (ARRG) systems, as it enables verification of how visual evidence supports generated reports. The design of the visual encoder largely constrains the explainability mechanisms that can be applied. Convolutional neural network (CNN) based encoders, for example, support gradient-based attribution methods such as Grad-CAM, allowing radiologists to localize image regions associated with specific report findings (Selvaraju et al. 2017). Such region-level explanations are particularly valuable for validating abnormalities such as pulmonary opacities or fractures. More advanced encoders incorporating multi-scale representations and attention mechanisms further enhance visual grounding by providing interpretability across anatomical hierarchies. Nevertheless, the growing complexity of deep learning models has intensified concerns regarding their “black-box” nature (Adadi and Berrada 2018; Budhkar et al. 2025), motivating the adoption of Explainable AI (XAI) techniques in ARRG pipelines. Within the context of feature extraction, explainability methods primarily emphasize region-level transparency, including saliency maps, attention visualization, Score-CAM (Wang et al. 2020), and related techniques that clarify image-text associations. Emerging approaches also explore feature-level interpretability, where visual representations are aligned with clinically meaningful concepts to improve semantic consistency (Yang et al. 2020). While methods such as SHAP (Lundberg and Lee 2017) and LIME (Ribeiro et al. 2016) are more commonly applied at later reasoning stages, their use highlights the increasing interest in quantifying feature contributions to diagnostic outputs. By enabling visual validation of encoder behaviour, these explainability mechanisms support error detection, strengthen radiologist trust, and form a critical foundation for clinically deployable ARRG systems (Figure 3).

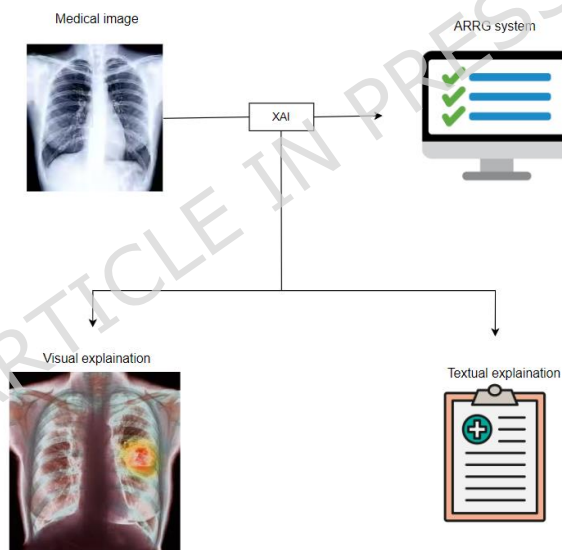


Figure 3: Explainability mechanisms in ARRG systems.

Domain-Specific Pre-training: Bridging the Semantic Gap

Domain shift between natural images and radiological scans limits the effectiveness of ImageNet pre-training. To address this, recent ARRG systems increasingly rely on medical-specific pre-training. CheXNet (Rajpurkar et al. 2017), MedCLIP (Z. Wang et al. 2022), BioViL, and PubMedCLIP improve visual semantic alignment and reduce false negatives for rare pathologies by 15–25% compared to generic initialization, directly enhancing clinical safety.

Scalability and Deployment Constraints

Encoder complexity introduces practical trade-offs between performance and deployability. While GPU-based systems meet real-time requirements in tertiary centers, CPU-only deployment remains challenging in resource-limited settings. Multi-view and multi-modal encoders (Jing et al. 2018b; Y. Liu et al. 2024) further improve reasoning across complementary inputs but increase computational cost by 2–3×, necessitating careful consideration for widespread adoption.

Alternative and Emerging Feature Learning Paradigms

Beyond supervised CNN encoders, alternative approaches such as GAN-based feature extraction (Liao et al. 2023) and self-supervised contrastive learning (Hou et al. 2023) address data annotation bottlenecks. While promising, these methods remain limited by training instability, interpretability challenges, and the absence of standardized clinical validation protocols key barriers to regulatory translation.

Reasoning, Knowledge Integration, and Report generating

Following visual feature extraction, the second stage of ARRG focuses on clinical reasoning and report drafting, where visual representations are transformed into structured, diagnostically meaningful narratives. This stage plays a central role in determining report coherence, terminology accuracy, and alignment with clinical knowledge. Unlike the feature extraction stage, which emphasizes visual grounding, Stage 2 addresses higher-level reasoning by integrating contextual information, medical knowledge, and language modeling to support clinically consistent interpretations. Recent ARRG systems increasingly augment this stage with explicit knowledge integration, retrieval mechanisms, and advanced generative models to reduce semantic ambiguity and improve diagnostic consistency (Huang and Li 2024; Rojas et al. 2025). As a result, reasoning and drafting modules have evolved from simple sequence decoders to knowledge-aware and retrieval-augmented architectures capable of leveraging external medical context. The following subsections examine these developments, focusing on how different reasoning paradigms influence clinical accuracy, transparency, and scalability in ARRG systems.

Graph-Enhanced Reasoning

To improve the semantic consistency and clinical validity of Automatic Radiology Report Generation (ARRG) systems, several studies have explored graph-based reasoning mechanisms, primarily through knowledge graphs (KGs) and graph neural networks (GNNs). These approaches explicitly model relationships among clinical entities and have been introduced to address the limitations of purely sequential or attention-based reasoning in complex radiological narratives (Gu et al. 2024). Knowledge graphs represent medical concepts as nodes and their semantic relationships as edges, providing a structured representation of domain knowledge (Figure 3). As defined by (Y. Zhang et al. 2020), a knowledge graph is a semantic network in which entities or concepts are connected through directed edges encoding meaningful relations such as similarity or clinical association. In ARRG, KGs are typically integrated during the report drafting stage to constrain generation toward medically consistent terminology and relationships. For instance, (Su et al. 2026) introduced a high-information-density knowledge injection module that incorporates medical domain knowledge into the generation process, resulting in reports that are more clinically aligned and diagnostically accurate.

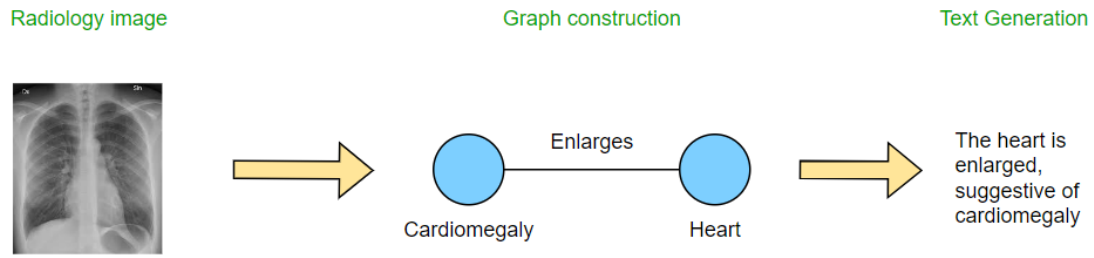


Figure 4: Image - graph - text flow

Graph neural networks complement knowledge graphs by enabling learnable relational reasoning through message passing and node aggregation. Unlike static KGs, GNNs can dynamically update node representations based on contextual information. In ARRG, models such as Graph Convolutional Networks (GCNs) and Graph Transformers (GTRs) have been trained on radiological ontologies derived from image-report pairs (Liao et al. 2023). Early applications by (Li et al. 2019) and (Y. Zhang et al. 2020) demonstrated that incorporating graph structures improves semantic coherence in generated reports, particularly for disease-finding associations. However, initial graph-based ARRG models were limited by coarse disease taxonomies and minimal interpretability, often relying on a restricted set of predefined clinical nodes. For example, (Y. Zhang et al. 2020) constructed disease-centric graphs for chest imaging, while (Li et al. 2019) combined graph reasoning with template retrieval based on detected abnormalities.

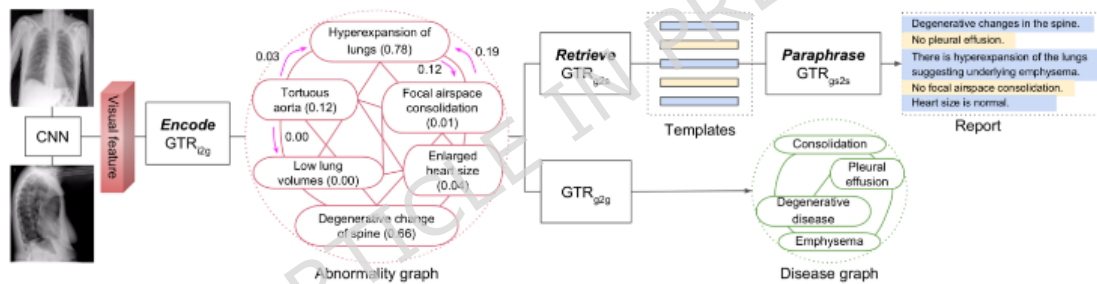


Figure 5: ARRG system architecture using knowledge-driven encode, retrieve and paraphrase module (Sloan et al. 2024)

More recent work has expanded this paradigm. (Yin et al. 2024) proposed Graph2Text, leveraging multiple graph domains including meaning representations and external knowledge graphs to generate more fluent and contextually rich reports. In a related effort, DKA-RG (Yin et al. 2024) integrates global and disease-level alignment to better capture fine-grained pathological characteristics, achieving improved report quality. Overall, graph-enhanced reasoning introduces an explicit relational layer between visual features and language generation, supporting more structured clinical reasoning in ARRG systems (Figure 4). While challenges remain regarding scalability and standardization of medical knowledge sources, graph-based models represent a promising direction for improving diagnostic consistency and semantic grounding in automated radiology reporting.

Retrieval-Augmented Generation and Agentic AI in Clinical Radiology

As Automatic Radiology Report Generation (ARRG) systems have evolved from sequential decoders and transformer-based architectures toward large language models (LLMs), limitations in factual grounding, rare-pathology coverage, and hallucination control have become increasingly apparent. To address these challenges, Retrieval-Augmented Generation (RAG) and Agentic AI have emerged as complementary reasoning paradigms that extend generative models with external knowledge access and multi-step decision-making. Together, they mark a transition from purely generative ARRG systems toward more context-aware and clinically grounded architectures.

- Retrieval-Augmented Generation in ARRG

RAG enhances the reliability of automated radiology reports by coupling LLM-based generation with external knowledge retrieval. In this paradigm, image-derived findings or intermediate representations are transformed into queries that retrieve relevant clinical information from curated sources such as PubMed, UMLS, Radiopaedia, or large-scale radiology corpora (Fink et al. 2025; Weinert and Rauschecker 2025). The retrieved content is then used to condition report generation, effectively acting as an external memory that constrains output toward verifiable medical knowledge. Several RAG-based ARRG systems have demonstrated improvements in factual consistency and diagnostic relevance. RadioRAG (Arasteh et al. 2025), which integrates real-time retrieval from Radiopaedia, reported notable gains in diagnostic accuracy for selected subspecialties.

(Sun et al. 2025)System	Knowledge Source	Retrieval Method	Clinical Performance	Deployment Status
RadioRAG (Arasteh et al. 2025)	Radiopaedia (real-time)	Dense retrieval (FAISS)	+54% relative diagnostic accuracy; matched/exceeded radiologists	Research prototype
MMed-RAG (Xia et al. 2025)	MIMIC-CXR, PubMed	Universal multimodal retriever	Superior factual completeness; +2–6.5% over baseline	Research prototype
FactMM-RAG (Sun et al. 2025)	RadGraph, multimodal corpora	RadGraph-guided retrieval	+2% RadGraph F1; +6.5% factual correctness	Research prototype
BioGPT-RAG	PubMed abstracts	Sparse retrieval (BM25)	Reduced factual inconsistencies by 18%	Early-stage research

Table 5: Key RAG Models and Performance in ARRG

MMed-RAG (Xia et al. 2025) combines MIMIC-CXR reports with PubMed literature to enhance report completeness, while FactMM-RAG (Sun et al. 2025) employs RadGraph-guided retrieval to improve structural and factual alignment. BioGPT-RAG further illustrates the value of literature-grounded generation by reducing factual inconsistencies through targeted PubMed retrieval. A comparative overview of representative RAG-based ARRG systems is provided in Table 5. Despite these advantages, RAG introduces new challenges for clinical deployment. Retrieval latency, dependency on the quality and relevance of external documents, and the need for continuous knowledge-base maintenance complicate integration into time-sensitive workflows. Moreover, while retrieval can reduce hallucinations, it does not eliminate them, particularly when retrieved sources are incomplete or contradictory. Consequently, current RAG-based ARRG systems are best positioned as assistive tools, supporting radiologists through retrieval-informed suggestions rather than autonomous report generation.

- Agentic AI: Toward Autonomous Reasoning

Agentic AI extends the RAG paradigm by introducing multi-step reasoning, tool use, and decision-making into ARRG workflows (Khosravi et al. 2026). Rather than relying on a single retrieval-and-generate cycle, agentic systems decompose report generation into sequential tasks, such as image analysis, retrieval of prior studies or guidelines, differential diagnosis generation, consistency checking, and final report drafting. Frameworks such as AutoGen, LangChain/LangGraph, and Med-Gemini exemplify this approach by enabling LLMs to dynamically invoke tools and coordinate complex reasoning processes (Figure 6).

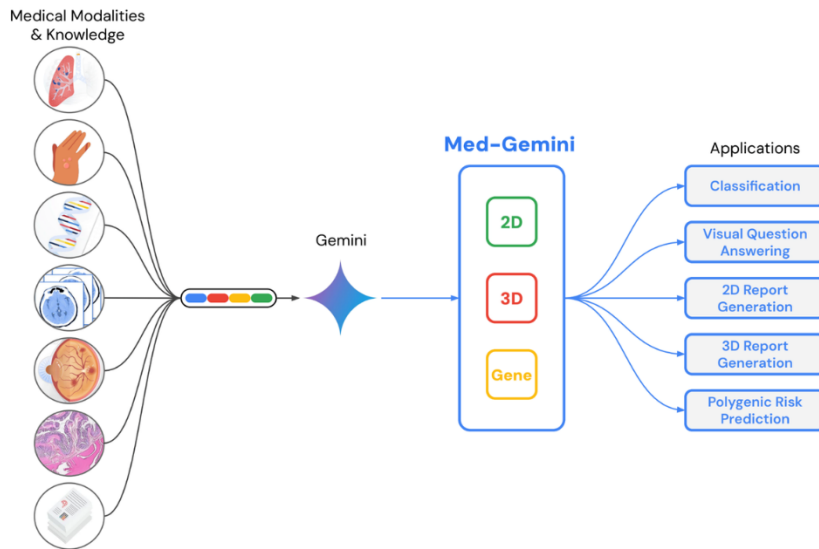


Figure 6: Med-Gemini⁴, a family of Gemini models fine-tuned for multimodal medical domain applications

In a conceptual agentic ARRg pipeline, a system may ingest a chest X-ray, access relevant electronic health record data, retrieve clinical guidelines (e.g., Fleischner Society recommendations), generate and refine a differential diagnosis, and perform internal consistency checks before producing a structured report. This workflow more closely mirrors human radiological reasoning and aims to move ARRg beyond static text generation toward context-aware diagnostic assistance (Figure 6).

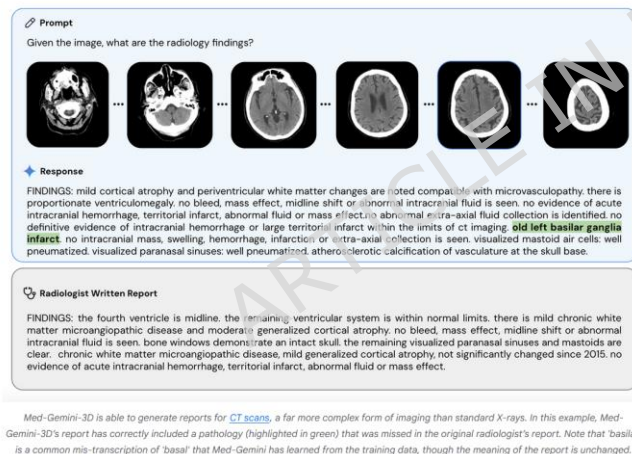


Figure 7: Med-Gemini-3D⁵ report from CT scan compared with radiologist written report

However, Agentic AI remains largely exploratory in clinical radiology. Multi-step pipelines compound error risk, introduce substantial latency, and pose significant challenges for explainability, as tracing the influence of individual tools or reasoning steps is non-trivial (Salehi et al. 2025). In addition, regulatory and governance frameworks are not yet equipped to accommodate semi-autonomous decision-making systems, and the associated computational and infrastructure requirements limit deployment to well-resourced academic settings.

- Comparative Perspective: RAG vs. Agentic AI

⁴ <https://research.google/blog/advancing-medical-ai-with-med-gemini/>

⁵ <https://research.google/blog/advancing-medical-ai-with-med-gemini/>

From a clinical readiness standpoint, RAG and Agentic AI follow distinct but complementary trajectories. RAG represents a near-term, deployable enhancement to ARRG, offering improved factual grounding and rare-disease coverage with manageable integration complexity.

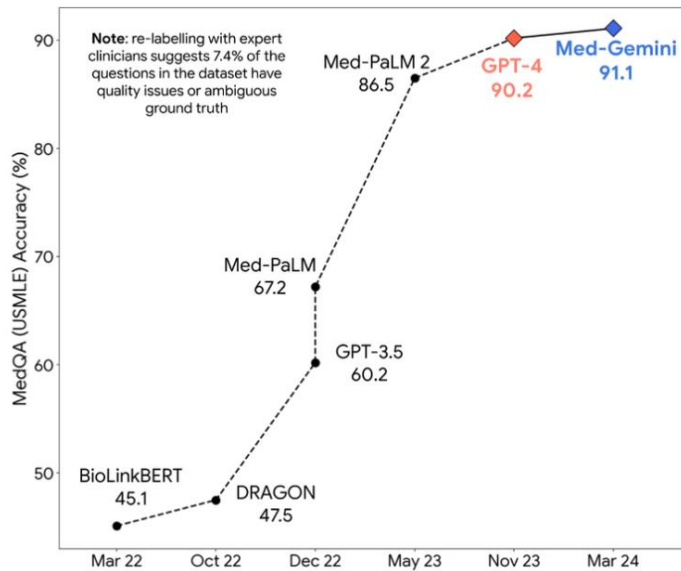


Figure 8: Med-Gemini Vs previous RAG models

In contrast, Agentic AI reflects a long-term vision for autonomous, human-like clinical reasoning, but faces substantial barriers related to reliability, explainability, latency, governance, and cost. A comparative assessment of these approaches is summarized in Table 6 and illustrated in Figure 8.

Dimension	RAG	Agentic AI
Technical Maturity	Medium (active research, multiple prototypes)	Low (early proof-of-concept only)
Clinical Accuracy	Moderate improvement (2–6.5%)	Unknown
Latency	3–10 s	15–40 s
Explainability	Poor	Very Poor
Deployment Cost	\$50K–\$200K	\$500K–\$2M+
Regulatory Path	Extension of existing AI frameworks	Requires new governance paradigms
Realistic Deployment	2027–2030 (narrow domains)	2032+ (if ever)
Recommended Clinical Role	Assistive tool for radiologist review	Research only

Table 6: Comparative Assessment of RAG and Agentic AI for ARRG

Together, RAG and Agentic AI outline a developmental pathway for ARRG systems: incremental improvements in clinical support and report accuracy in the near term, alongside ongoing exploration of more autonomous reasoning paradigms over the longer horizon.

Report Generation Models: From Decoders to Clinical Language Models

Once visual features are extracted and contextual reasoning is performed, Automatic Radiology Report Generation (ARRG) systems must translate internal representations into clinically accurate and structured natural language. This stage is not merely linguistic generation but a safety-critical component, as omissions, hallucinations, or miscalibrated uncertainty can directly influence diagnostic decisions. Over time, report generation models have evolved from sequential decoders to large-scale clinical language models, each introducing distinct trade-offs in accuracy, interpretability, and deployment feasibility.

- LSTM and GRU based Decoders

Early ARRG systems predominantly employed recurrent neural networks (RNNs), particularly LSTM and GRU architectures, due to their ability to model sequential dependencies in clinical text (Das et al. 2023). These models decode image-derived features word by word, capturing report structure while maintaining temporal coherence. LSTM-based decoders were widely adopted between 2015 and 2020 (Xue et al. 2018; Li et al. 2018) and achieved moderate performance on benchmark datasets such as IU X-Ray, with reported BLEU-4, CheXbert, and RadGraph F1 scores reflecting reasonable fluency but limited clinical depth. A key limitation of these models was their tendency to overproduce “normal findings,” particularly in cases involving rare or subtle pathologies, resulting in safety-critical false negatives. Attention mechanisms partially mitigated this issue by providing coarse visual grounding (Jing et al. 2018b), although such explanations reflected correlation rather than causal reasoning. GRU-based decoders demonstrated comparable clinical performance while offering improved computational efficiency, making them attractive for resource-constrained settings such as rural hospitals or edge deployments (Chen et al. 2020).

- Transformer-Based Decoders: Structured Reasoning at Scale

Transformer-based decoders represented a major advance by replacing recurrence with self- and cross-attention mechanisms, enabling parallel decoding and improved modeling of long-range dependencies (Chen et al. 2020; Nooralahzadeh et al. 2021; Alfarghaly et al. 2021). These architectures consistently demonstrated improved clinical metrics, including higher CheXbert and RadGraph F1 scores, and produced more coherent multi-section reports, particularly for multifocal or bilateral abnormalities. From a clinical perspective, multi-head attention provides richer visual-textual alignment than RNN-based attention, though the interpretability of these mechanisms remains complex and inconsistent across cases. Transformer decoders are also substantially larger than RNNs, with increased memory requirements and inference latency, which can limit deployment in settings without dedicated GPU infrastructure.

- Large Language Models: Linguistic Fluency versus Clinical Reliability

More recent work has explored large language models (LLMs) such as GPT-4, LLaMA, and PaLM, adapted or fine-tuned for medical and multimodal applications (M. Li, Lin, et al. 2023; Nakaura et al. 2024). These models exhibit exceptional linguistic fluency, broader rare-disease coverage, and zero-shot generalization capabilities. However, reported evaluations indicate persistent hallucination and hedging behaviour, as well as limited transparency, with natural-language explanations often reflecting post-hoc rationalizations rather than explicit reasoning. From a deployment standpoint, LLM-based ARRG systems face significant scalability, latency, and governance challenges due to model size, inference cost, and data privacy constraints. As a result, their use in radiology remains largely experimental and unsuitable for unsupervised clinical deployment.

- Comparative Overview and Deployment Considerations

No single report generation architecture fully satisfies all clinical requirements. RNN-based decoders remain practical for high-volume, low-complexity workflows, transformers offer the most balanced performance for structured reporting in well-resourced environments, and LLMs provide powerful research tools for reasoning and rare-disease exploration.

Architecture	Clinical Accuracy (F1)	Explainability	Inference Speed	Deployment Readiness	Best Use Case
LSTM	Moderate (0.42–0.53)	Good (attention maps)	Fast (80–120ms)	High	Routine chest X-rays, high-volume screening
GRU	Moderate (0.40–0.51)	Good (attention maps)	Very Fast (60–90ms)	High	Rural hospitals, mobile radiology
Transformer	High (0.51–0.61)	Moderate (multi-head attention, complex)	Moderate (150–250ms)	Medium	Complex cases, multi-section reports, academic centers
LLM	Variable (0.41–0.48)	Poor (post-hoc explanations)	Slow (2–8s)	Low	Rare diseases, multi-modal reasoning, research applications

Table 7: Comparative Overview of ARRG Report Generation Architectures

Across all architectures, persistent translational gaps include hallucination control, calibrated uncertainty estimation, longitudinal reasoning across exams, and robust multi-institutional validation (Table 7). At present, evidence supporting autonomous clinical deployment remains limited, underscoring the continued need for human oversight.

Enhancement, Refinement, and Explainability

While feature extraction and report drafting determine the core content of Automatic Radiology Report Generation (ARRG) systems, a final enhancement stage is essential to improve clinical reliability, interpretability, and safety. This stage focuses on refining generated outputs, correcting structural or semantic inconsistencies, and providing explanations that enable radiologist oversight. Recent ARRG architectures increasingly incorporate attention mechanisms, reinforcement learning strategies, and generative AI models to address these requirements.

- Attention Mechanisms for Refinement and Visual–Textual Alignment

Attention mechanisms, first introduced by (Bahdanau et al. 2014), have become a foundational component in ARRG pipelines by enabling selective focus on relevant parts of visual and textual representations. Attention serves as an interface between encoders and decoders, allowing the generation process to dynamically weight informative image regions or semantic tokens. Comprehensive overviews of attention variants and their learning dynamics are provided by (Niu et al. 2021) and (Brauwers and Frasincar 2022). In ARRG systems, attention is employed not only to enhance feature extraction (Stage 1) but also to refine report content by strengthening alignment between image evidence and generated statements. Encoder-decoder frameworks augmented with attention, such as those proposed by (Xue et al. 2018), demonstrate improved localization of abnormalities and

reduced omission of salient findings. More recent co-attention and cross-modal attention mechanisms further support interpretability by jointly modeling interactions between image regions and clinical terminology, facilitating post-hoc verification by radiologists.

- Generative AI Models for Report Refinement and Interaction

Beyond architectural attention mechanisms, generative AI models have been introduced to support higher-level refinement, long-context reasoning, and interactive correction of generated reports. Table 8 summarizes representative generative AI models applied in ARRG, spanning multimodal foundation models (e.g., Med-Gemini, Med-PaLM), domain-adapted language models (BioGPT, ChestBioX-Gen), and interactive systems (RaDialog).

These models extend traditional report generation by enabling iterative refinement, contextual consistency across long reports, and, in some cases, dialogue-based interaction with clinicians. For example, ChestBioX-Gen combines BioGPT with co-attention mechanisms to improve diagnostic precision in chest X-ray reporting, while RaDialog supports collaborative clarification of findings through conversational interfaces. Although such systems demonstrate promising performance, their primary contribution at this stage lies in post-generation enhancement rather than autonomous diagnosis

Model	Developer	Input Modalities	Key Features	Primary Use Cases	Notable Performance
Med-Gemini	Google	2D & 3D medical images, text	Multimodal integration; long-context reasoning; capable of generating reports from complex imaging modalities like CT scans.	Radiology report generation; visual question answering; anomaly detection.	Demonstrated ability to generate accurate reports for CT scans, sometimes identifying findings missed by radiologists.
Med-PaLM	Google	Text, medical images	Multimodal generative model; integrates clinical language and imaging data; designed for various biomedical tasks.	Clinical question answering; radiology report generation.	Achieved expert-level performance on USMLE-style questions; effective in synthesizing information from diverse medical data.
BioGPT	Microsoft	Text	Pretrained biomedical language model; focuses on understanding and generating biomedical text.	Biomedical text generation; information retrieval.	Excels in generating coherent biomedical text; forms the basis for models like ChestBioX-Gen for radiology reports.
ChestBioX-Gen	Université de Moncton	Chest X-ray images, text	Combines BioGPT with co-attention mechanisms; uses CheXNet for feature extraction from images.	Chest X-ray report generation.	Enhances precision in diagnostic outcomes; reduces radiologist workload by automating report generation.
MiniGPT-Med	KAUST	Medical images, text	Vision-language model; capable of report generation, visual question answering, and disease identification.	Radiology diagnosis assistance; report generation.	Achieved state-of-the-art performance in medical report generation; improved diagnostic accuracy across imaging modalities.

RaDialog	Technical University of Munich	Medical images, text	Integrates visual features with structured pathology findings; supports interactive dialogue.	Radiology report generation; conversational assistance.	Provides clinically correct reports; facilitates collaborative diagnostic processes through interactive capabilities.
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Table 8: Overview of generative AI Models in ARRГ

- Reinforcement Learning for Output Optimization

Reinforcement learning (RL) has also been explored as a refinement strategy for ARRГ systems, particularly to optimize long-sequence generation where conventional likelihood-based training may not align with clinical evaluation metrics. In RL-based ARRГ, the generation process is framed as a sequential decision-making problem, where rewards reflect report-level objectives such as linguistic fluency or clinical correctness. Li et al. (2018a) demonstrated the use of policy gradient methods to optimize hierarchical report generation by directly incorporating evaluation metrics as reward signals. This approach allows models to adjust generation behavior toward clinically meaningful outcomes, addressing exposure bias and improving global coherence. While RL can enhance report quality, its sensitivity to reward design and training instability limits widespread adoption in clinical settings.

- Explainability and Trust Calibration

A central objective of this stage is to bridge the gap between automated generation and clinical trust. Attention visualizations, alignment maps, and interactive refinement interfaces provide mechanisms for radiologists to audit generated reports, detect errors, and understand model behaviour. However, explainability remains uneven across ARRГ systems, particularly for large generative models where post-hoc explanations may not reflect internal reasoning processes (Dietrich 2025; Su et al. 2026). Overall, enhancement and refinement mechanisms serve as a critical safeguard layer in ARRГ pipelines, improving transparency, consistency, and user trust. By supporting verification and iterative correction rather than autonomous decision-making, this stage plays a pivotal role in preparing ARRГ systems for real-world clinical integration and directly informs the evaluation criteria discussed in the following section.

Quantitative Synthesis: Architectures, Data, and Evaluation

This section consolidates quantitative evidence across ARRГ studies, linking architectural evolution, dataset characteristics, and evaluation practices to clinical performance and translational readiness. Rather than reporting isolated metrics, the analysis emphasizes how architectural choices, data scale, and evaluation design collectively shape clinical utility, reproducibility, and deployment feasibility.

Evolution of AI Architectures in ARRГ

ARRГ architectures have evolved in response to growing clinical demands for accuracy, coherence, and scalability. **Table 8** summarizes this evolution, highlighting four dominant architectural generations. Early CNN–RNN systems (2015–2018) paired convolutional encoders with LSTM or GRU decoders, achieving proof-of-concept performance with relatively simple pipelines. Subsequent attention-enhanced RNN models (2018–2020) improved visual–text alignment and abnormality description, addressing early failures in lesion localization and sentence coherence (Jing et al. 2018b; Xue et al. 2018).

Generation	Period	Modality	Image Encoder	Text Decoder	Attention	Representative Models	Key Strengths
CNN–RNN	2015–2018	Image	CNN (VGG)	LSTM/GRU	Optional	(Xu et al. 2016)	Simplicity, feasibility

Attention–RNN	2018–2020	Image	CNN	Attn-LSTM	Yes	(Jing et al. 2018b; Xue et al. 2018)	Better alignment
Transformer	2020–2023	Image	CNN / ViT	Transformer	Multi-head	(Chen et al. 2020; Liu et al. 2021)	Long-range modeling
Generative / LLM	2023–Now	Multimodal	ViT, CLIP	GPT, PaLM, LLaVA	Yes	Med-PaLM;	Multimodal reasoning

Table 9: AI Architecture Evolution in ARRG

From 2020 onward, transformer-based decoders introduced parallel decoding and long-range dependency modeling, yielding consistent gains in both linguistic and clinical metrics (Chen et al. 2020; Liu et al. 2021). Most recently, generative and large multimodal language models (2023–present) such as Med-PaLM and LLaVA-Med extend ARRG to multimodal reasoning, external knowledge integration, and zero-shot generalization (Tu et al. 2023). Vision–Language Models (VLMs), including Flamingo (Alayrac et al. 2022), GIT (J. Wang et al. 2022), and LLaVA-Med (C. Li et al. 2023), represent a convergence point in this evolution, directly aligning with the Stage 2 reasoning and Stage 3 refinement mechanisms discussed in Section 3.

Datasets for Training and Benchmarking

ARRG performance is strongly conditioned on dataset scale, annotation quality, and modality coverage. Public datasets have enabled rapid progress, but also introduced systematic biases toward chest X-ray imaging and adult populations. Early ARRG research relied on OpenI and IU X-ray datasets (Demner-Fushman et al. 2015), which provided paired images and reports but limited diversity.

Dataset	Images	Reports			
	Image modality & body part (nO. images)	Format	Type	number	Report format
IU X-ray (Demner-Fushman et al. 2015)	Chest (lateral and frontal view) X-rays (n=7, 470)	JPEG, DICOM	Semi Structured reports + MeSH tags	3,995	TXT
ChestX-ray8 (Wang, Peng, Lu, Lu, Bagheri, and R. M. Summers 2017)	Chest (frontal view) X-ray (n=108,948)	DICOM	Semi structured + labels	NA ⁶	TXT
ChestX-ray14 (Wang, Peng, Lu, Lu, Bagheri, and R. Summers 2017)	Chest (frontal view) X-rays (n=112,120)	DICOM	Semi structured + labels	NA	NA
CheXpert (Irvin et al. 2019)	Chest (frontal and lateral) X-ray (n=224,316)	NA	Free text reports + labels	NA	TXT
MIMIC-CXR (Johnson et al. 2019)	Chest (Frontal and lateral views) X-ray (n=377,110)	JPEG, DICOM	Semi Structured report + labels	227,835	TXT

⁶ Not Available

PadChest (Bustos et al. 2020)	Chest (Frontal CXR) X-ray (n=160,000)	DICOM	Semi-structured	(Spanish partial English) +	TXT
VinDr-CXR (Nguyen et al. 2022)	Chest (Frontal CXR) X-ray (n=18,000)	PNG	NA	NA	CSV (annotations)
MIMIC-IV (Johnson et al. 2023)	Chest, Abdomen, CT, X-ray	DICOM	Unstructured	500,000	TXT

Table 10: Datasets used in ARRG systems

The introduction of MIMIC-CXR (Johnson et al., 2019) marked a turning point, offering over 370,000 images and 227,000 reports, enabling transformer and LLM-based models to learn clinically meaningful distributions. Table 9 summarizes the most commonly used datasets. Despite recent expansions (e.g., MIMIC-IV), multimodal and non-X-ray datasets remain scarce, constraining generalization to MRI, CT, and PET imaging.

Imaging Modality Constraints: X-ray versus CT, MRI, and PET

As summarized in Table 10, ARRG systems demonstrate substantially stronger performance on chest X-rays than on volumetric or functional imaging modalities. This disparity reflects differences in image dimensionality, diagnostic complexity, data availability, and regulatory risk.

Dimension	X-ray (2D Projection Imaging)	CT (Volumetric Imaging)	MRI (Multi-parametric Volumetric Imaging)	PET (Functional Imaging)
Image Dimensionality	Single 2D image	3D volume (hundreds of slices)	3D volume with multiple sequences	3D volume + temporal uptake data
Diagnostic Complexity	Relatively constrained (e.g., pneumonia, effusion, cardiomegaly)	High anatomical detail; subtle spatial relationships	High tissue contrast; sequence-dependent interpretation	Functional/metabolic interpretation with low spatial resolution
Reasoning Requirements	Local pattern recognition	Multi-slice spatial aggregation	Cross-sequence and anatomical-functional reasoning	Temporal, quantitative, and anatomical integration
Clinical Context Dependence	Limited	Moderate (contrast phase, protocol)	High (indication, sequence choice, prior exams)	Very high (tracer type, uptake timing, oncology history)
Data Availability	Large, publicly available labeled datasets	Moderate; fewer large-scale annotated datasets	Limited; heterogeneous protocols and labels	Very limited; sparse labeled datasets
Annotation Cost	Low	High	Very high	Very high
Report Structure Variability	Relatively standardized	Moderate variability	High variability	High variability
ARRG Model Performance	Strong and stable	Moderate, inconsistent	Weak to moderate	Weak

Regulatory Risk	Lower (simpler diagnostic scope)	Higher	Higher	Highest
Clinical Deployment Status	Limited real-world deployment achieved	Primarily experimental	Largely pre-clinical	Research-only

Table 11: Imaging Modality Constraints Affecting ARRГ System Performance

X-rays benefit from abundant labeled datasets and relatively constrained diagnostic vocabularies, enabling earlier regulatory approvals and limited clinical deployment. In contrast, MRI, CT, and PET require longitudinal reasoning, multi-slice aggregation, and integration of clinical context capabilities that remain underdeveloped in current ARRГ architectures.

Evaluation Metrics: From Linguistic Fluency to Clinical Efficacy

ARRГ evaluation practices reveal a critical gap between linguistic quality and clinical safety. As shown in Table 11, most studies continue to rely heavily on NLP-based metrics (BLEU, ROUGE, METEOR, CIDEr), which correlate weakly with diagnostic correctness (Kaur et al. 2022). In response, clinical efficacy metrics (CEMs) such as Clinical Accuracy (Liu et al. 2019), RadGraph F1, CheXbert similarity, and radiologist scoring have gained prominence. These metrics better capture factual correctness, pathology presence, and report-level safety but are costlier and less standardized.

Metric	Purpose	Algorithm	Strengths	Weaknesses
NLP-Based Metrics				
BLEU (Papineni et al. 2002)	Machine Translation	Modified n-gram precision + brevity penalty	Standardized, fast	Poor semantic correlation
ROUGE (Lin 2004)	Summarization/NLG	Overlap-based recall (ROUGE-N), LCS (ROUGE-L)	Commonly used, recall-based	Insensitive to semantics
METEOR (Banerjee and Lavie 2005)	Machine Translation	Word alignment + synonym/stem + F1 + penalty	Captures basic semantic equivalence	Computationally heavier than BLEU/ROUGE
CIDEr (Vedantam et al. 2014)	Image Captioning	TF-IDF weighting of n-grams against reference set	Captures consensus and salience	Biased toward frequent phrases, not semantic
SPICE (Anderson et al. 2016)	Image Captioning	Converts captions into tuples (objects, attributes, relations) and compares	Captures semantic propositions	Dependent on parsing accuracy, slow
BERTScore (T. Zhang et al. 2020)	Sentence Similarity/NLG	Cosine similarity of BERT embeddings (precision/recall/F1)	Strong semantic understanding	No clinical knowledge, slower
CLIPScore (Hessel et al. 2022)	Image Captioning/VQA	Cosine similarity of CLIP's image and text encodings	Vision-language alignment	Not radiology-specific
Clinical Relevance Metrics				

MeSH Accuracy (MA) (Xin et al. 2019)	Indexing/Information Retrieval	Manual binary or scalar correctness review	Interpretable, uses standard ontology	Depends on accurate MeSH annotation
Clinical Accuracy (CA) (Liu et al. 2019)	Clinical Fact Evaluation	Structured comparison of RadGraph annotations	Focuses on safety and correctness	Costly to evaluate at scale
RadGraph Score	Clinical Fact Extraction	Structured comparison of RadGraph annotations	Clinically grounded, interpretable	Requires gold standard RadGraph parsing
CheXbert Similarity (Picha et al. 2024)	Clinical Labeling	Classification agreement on disease labels using CheXbert	Easy to compute, clinically relevant	Limited taxonomy (only 14 labels)
Radiologist Score	Clinical Judgment	Manual scoring, Likert or pairwise ranking	Gold standard, trust-based	Expensive, subjective, not reproducible

Table 12: Evaluation metrics used in ARRG

Crucially, no single metric adequately reflects real-world clinical risk. High BLEU scores may coexist with missed critical findings, underscoring the need for multi-metric evaluation frameworks combining linguistic fluency, factual grounding, and expert judgment.

Comparative Analysis of Key Studies

Table 12 synthesizes representative ARRG studies across architectures, datasets, and evaluation strategies. The analysis reveals three consistent trends:

1. Architectural sophistication improves average metrics but does not eliminate hallucination or omission errors.
2. Larger datasets yield better generalization, yet institutional bias persists.
3. Human evaluation remains indispensable, particularly for safety-critical findings.

Despite steady progress, no ARRG system has demonstrated non-inferiority to radiologists in prospective, multi-center clinical trials highlighting a persistent translational gap that motivates the discussion in Section 5.

Articles	Architecture	Algorithms	Datasets (size) / body part	Strength	limitations
(Zhang et al. 2017)	Encoder-decoder	CNN-RNN-Att_mecha	BCIDR (bladder cancer images and its diagnostic reports. 1,000 images RVB 500x500 were collected) / bladder, two public CIFAR ('CIFAR-10 and CIFAR-100 Datasets', n.d.) datasets are used to test the CNN part of the model / naturel images	-First interpretable attention-based model -Unified network architecture -Direct mapping medical images-reports -Network attention visualization -Specific optimization strategies to prevent overfitting	-Small dataset -The need of effective regularization techniques -Complexity -Dependence on Report Quality
(Xue et al. 2018)	Encoder-decoder	CNNs-Bi_LSTM (hierarchical) & attention mechanism	IU X-ray (Demner-Fushman et al. 2015) 7,470 chest X-ray images and 3,955 radiology reports / Chest	-Generation of long and detailed reports -Good result shown by the model	-Data missing -The need of large dataset -Evaluation metrics which consider keyword accuracy and grammar correctness
(Jing et al. 2018c)	Encoder-decoder	CNNs-Hierarchical LSTM & co-attention mechanism	IU X-ray (Demner-Fushman et al. 2015) 7,470 chest X-ray images and 3,955 radiology reports / Chest PEIR Gross ('PEIR Digital Library', n.d.) (7,442 image-caption pairs from 21 different sub-categories) / Chest	-Exploration of visual and semantic information -Capture long-range semantics and produce high quality long texts	

(Li et al. 2019)	Retrieval	CNNs & GTRs	IU X-ray (Demner-Fushman et al. 2015) 7,470 chest X-ray images and 3,955 radiology reports / Chest CX-CHR (Li et al. 2019) private dataset (35,609 patients and 45,598 chest X-ray images with corresponding Chinese report) / Chest	-Generate accurate reports -Achieving the state-of-the-art results on two medical report benchmarks, -Good medical abnormality and disease classification accuracy	Complexity
(Gale et al. 2019)	Encoder-decoder	CNN-SRNN-Att (DenseNet (Huang et al. 2017) , LSTM-Att_mech)	X-rays (Gale et al. 2017) (50,363 frontal pelvic X-rays, containing 4,010 hip fractures) / Pelvic	-Easy to expand -Bridge the gap between machine learning specialists and end-users -Produce high-quality 'human-style' explanations for diagnostic decisions in medical imaging tasks	-May not capture all nuances of complex decisions -The need of more large datasets -Test the model with some large datasets used in ARR-G
(Syeda-Mahmood et al. 2020)	Retrieval	imageNet-pretrained, Feature Pyramid Network (Lin et al. 2016), VGGNet (Simonyan and Zisserman 2014)(16 layers) and ResNet (He et al. 2016)(50 layers)	MIMIC-4 (Johnson et al. 2019)(220,000 reports with associated frontal images) / Chest Indiana reports (2964 unique Indiana reports with associated images) / Chest NIH (Wang, Peng, Lu, Lu, Bagheri, and R. M. Summers 2017)(10,000 released images re-read by radiologists team) / Chest	-Learning fine-grained descriptions of findings from chest X-ray images -Using patterns of occurrences to retrieve and customize similar reports from a large database -DL network that recognizes both coarse and	-Dependence on Large Datasets -Negation detection complexity for the semantic accuracy -Restriction to generate diverse reports

			total of 232,964 image-report pairs	fine-grained descriptions of findings	
(Y. Zhang et al. 2020)	Encoder-decoder	Graph with Prior Knowledge CNN-GCN-att_mecha-LSTM	IU X-ray (Demner-Fushman et al. 2015) 7,470 chest X-ray images and 3,955 radiology reports / Chest	-Proposes a new evaluation metric - Decoder consist of topic/word level LSTM	- pre-constructed graph - The model may not be generalized on other datasets
(Alfarghaly et al. 2021)	Encoder-decoder	Chexnet (Chou [2018] 2024), CNN-BERT	IU X-ray (Demner-Fushman et al. 2015) 7,470 chest X-ray images and 3,955 radiology reports / Chest	-Fast to train -Don't need the vocabulary specification	Information missing in generated reports
(Xue et al. 2022)	Encoder-decoder	CNN-RNN (GRU bi) & Transformers	MIMIC-CXR (Johnson et al. 2019) (473,057 chest X-ray images and 206,563 reports) / Chest, IU X-ray (Demner-Fushman et al. 2015) 7,470 chest X-ray images and 3,955 radiology reports / Chest	-Alleviates the data bias problem -Achieves good accuracy and fluency.	-Complexity of the model -Its low inference speed and the suitable selection of the auxiliary signal

(Hou et al. 2023)	Encoder-decoder	Pretrained CNN_ RestNet-101(He et al. 2015) IU-MKG based on (Y. Zhang et al. 2020) Supervised contrastive learning Transformer based meth for report generation	IU X-ray⁷ (Demner-Fushman et al. 2015) 7,470 chest X-ray images and 3,955 radiology reports / Chest	-Clearly identifying abnormal areas in medical images by using SCL⁸ -The use of IU-MKG⁹ to handle the lack of medical terms and the discovery and association of diseases in reports generation	-relationship between diseases/organs/tissues is not specific -Uses only the IU X-ray dataset to test -Complexity
(Yi et al. 2024)	Encoder-decoder	disease tag, ResNet-101¹⁰(He et al. 2015), Reinforcement Learning¹¹, transformer decoder	IU X-ray¹² (Demner-Fushman et al. 2015) 7,470 chest X-ray images and 3,955 radiology reports / Chest, MIMIC-CXR¹³ (Johnson et al. 2019) (473,057 chest X-ray images and 206,563 reports) / Chest	-Qualitative evaluation & compared with previous works - Reinforcement Learning combine with NLG metrics to evaluate the model	-Computational resources -Some improvement can be made in feature extraction

⁷ The dataset is splited into train/test/validation set for 70%/20%/10%

⁸ Supervised Contrastive Learning

⁹ IU Medical Knowledge Graph

¹⁰ ResNet-101 pre-trained on the ImageNet to extract 7*7*2048 image

¹¹ NLG metrics based Reinforcement Learning to offer appropriate guidance for automatic radiology report generation

¹² We split the training/validation/testing in a ratio of 7:1:2

¹³ We follow the official split

(Y. Wang et al. 2024)	Encoder-decoder	CNN_ResNet-101(He et al. 2015), AutoEncoder, Transformer (multi-head self-attention and a multi-head cross-attention)	COV-CTR 728 images (349 for COVID-19 and 379 for Non-COVID) IU X-ray (Demner-Fushman et al. 2015) 7,470 chest X-ray images and 3,955 radiology reports / Chest	- Explicitly quantify the inherent visual-textual uncertainties in the multi-modal task - incorporate an auxiliary AutoEncoder branch	-Computational needs -Relationships between the visual and textual only builds at the report level
(Zhang and Jiang 2024)	Encoder-decoder	frozen medical multimodal, ResNet(He et al. 2015), ViT (Dosovitskiy et al. 2020), MedCLIP's frozen encoder(Z. Wang et al. 2022)	MIMIC-CXR (Johnson et al. 2019) (473,057 chest X-ray images and 206,563 reports) / Chest	- Generation of region specific anatomical terms - radiologists can specify manually their areas of interest	-sentences errors during dataset construction -significant gap between the effectiveness of report generation and sentence generation
(Y. Liu et al. 2024)	Encoder-decoder	Pre-trained DenseNet-121(Huang et al. 2017), Dual GCN, FFARM ¹⁴ , transformer-base encoder, and transformer-based decoder	CheXpert (Irvin et al. 2019) 224,316 chest X-ray images / Chest, IU X-ray ¹⁵ (Demner-Fushman et al. 2015) 7,470 chest X-ray images and 3,955 radiology reports / Chest	-The integration of feature fusion and relationship mining component -The model was compared to some relevant work and shown higher scores on metrics used	-The study used one medical dataset to compare its results with previous -The model need to be evaluate with qualitative metrics

Table 13: Comparative Table of Key ARRG Studies

¹⁴ feature fusion and relationship mining module¹⁵ the IU X-ray dataset was partitioned into train, validation, and test sets in a 7:1:2 ratio

Discussion and Future Directions

Synthesis of Findings Through the Clinical Translation Lens

This review reveals a clear and consistent trajectory in Automatic Radiology Report Generation (ARRG): a transition from fluency-oriented text generation toward systems increasingly constrained by clinical safety, reliability, and trustworthiness. While early CNN–RNN and attention-based architectures demonstrated feasibility, recent transformer, RAG, and LLM-driven systems reflect an explicit shift toward clinical reasoning, external knowledge integration, and multimodal understanding. However, when evaluated against clinical translation criteria—diagnostic accuracy, explainability, scalability, and regulatory readiness—substantial gaps remain. Across architectures, performance gains in linguistic metrics do not reliably translate into improved clinical correctness. Hallucination, poor generalization across institutions, and opaque reasoning persist as dominant failure modes, particularly in LLM-based and retrieval-augmented systems. These findings suggest that architectural sophistication alone is insufficient; progress in ARRG must be pipeline-aware, addressing failure points at each stage from feature extraction to refinement and explanation.

Critical Barriers to Clinical Deployment

Despite rapid technical progress, our analysis identifies several persistent barriers that currently prevent ARRG systems from safe, large-scale clinical deployment.

- Hallucination and factual inconsistency remain the most critical challenges. Even state-of-the-art LLM- and RAG-based models generate clinically plausible but incorrect statements, posing unacceptable risks in diagnostic contexts (Nakaura et al. 2024; Biswas et al. 2024). While retrieval and knowledge graphs reduce hallucination rates, they do not eliminate them.
- Limited generalization is another major concern. Most ARRG systems are trained and evaluated on a small number of public datasets (primarily IU X-ray and MIMIC-CXR), leading to dataset-specific biases and degraded performance in unseen clinical environments.
- Explainability deficits undermine clinician trust. Although attention maps, graph reasoning, and XAI tools provide partial transparency, current explanations often reflect correlations rather than causal clinical reasoning (Borys et al. 2023; Budhkar et al. 2025).
- Scalability and infrastructure constraints further limit adoption. Transformer and LLM-based systems impose high computational, latency, and maintenance costs, restricting deployment to well-resourced academic centers.
- Privacy, governance, and liability issues remain unresolved, particularly for cloud-based LLMs and dynamic retrieval systems that complicate responsibility attribution.
- Finally, evaluation misalignment persists. Overreliance on NLP metrics (BLEU, ROUGE, CIDEr) obscures safety-critical errors, while clinical metrics remain fragmented and costly to apply at scale.

Architectural and Methodological Pathways Forward

Based on the reviewed evidence, several actionable research pathways emerge:

1. Pipeline-aware design
Future ARRG systems should be explicitly structured around the three-stage pipeline:
robust visual grounding → knowledge-aware reasoning → constrained refinement and explanation.

Hybrid models combining CNN/ViT encoders, graph-based reasoning, and controlled language generation offer the most promise.

2. **Knowledge-constrained generation**
Knowledge graphs, ontology-guided decoding, and structured retrieval (Sun et al., 2025) should act as hard constraints, not optional enhancements, particularly for abnormal findings and rare diseases.
3. **Clinically aligned evaluation frameworks**
Standardized evaluation protocols combining RadGraph-based metrics, uncertainty calibration, and radiologist-in-the-loop assessment are urgently needed to replace metric fragmentation.
4. **Multimodal and longitudinal reasoning**
Integrating imaging with patient history, laboratory data, and prior exams remains an open challenge. Multimodal fusion and temporal modeling are essential for clinically realistic report generation.
5. **Human–AI collaboration over autonomy**
In the near term, ARRG systems should be framed as decision-support tools, emphasizing transparency, controllability, and interactive refinement rather than autonomous reporting.
6. **Regulatory-aware development**
Future research must align with regulatory expectations, prioritizing traceability, auditability, and prospective validation over benchmark performance alone.

Limitations of This Review

This review has several limitations. First, it relies primarily on publicly available datasets and published evaluations, which may not reflect real-world clinical performance. Second, quantitative comparisons are constrained by heterogeneous evaluation protocols and inconsistent reporting. Third, the rapidly evolving nature of LLM-based ARRG means that some findings may become outdated as new models emerge. Nonetheless, by adopting a pipeline-based and clinically grounded taxonomy, this review provides a durable framework for interpreting ongoing advances in the field.

Conclusion

Automatic Radiology Report Generation (ARRG) has evolved from fluency-focused models to clinically aware systems integrating multimodal deep learning, transformers, large language models (LLMs), and knowledge graphs. This review provides a comprehensive, clinically oriented synthesis, highlighting how architectural choices, enhancement strategies, and evaluation frameworks impact diagnostic accuracy, explainability, and real-world deployment. Our quantitative analysis demonstrates that while transformers and LLMs substantially improve report detail and reasoning, persistent challenges hallucination, limited generalization, and opaque decision-making still constrain clinical adoption. By systematically examining datasets, evaluation metrics, and model pipelines, we identify gaps between technical performance and clinical readiness, including the need for standardized datasets, robust multi-institutional validation, and explainable AI mechanisms that foster radiologist trust. This work uniquely bridges technical synthesis with practical translation barriers, offering actionable insights for regulatory approval, human-in-the-loop workflows, and multimodal integration. ARRG systems are poised to enhance diagnostic efficiency and radiologist support, but achieving safe, scalable, and clinically deployable models requires coordinated advances in model design, evaluation, interpretability, and ethical implementation. This review provides a holistic roadmap, guiding future research toward patient-centered, trustworthy, and high-impact AI-assisted radiology.

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